These have larger river basins and a catchment area.	These have smaller river basins and a catchment area.
The deltas of the Himalayan rivers are very large.	The deltas present are small. Some rivers just have estuaries.
Himalayan rivers are more navigable in the plains.	These are less navigable.
The Himalayan drainage has many antecedent rivers.	In Peninsular drainage, there are only consequent rivers.
The drainage pattern is mainly dendritic.	The drainage pattern is mainly rectangular or trellis.

- Examples of antecedent rivers are Indus, Brahmaputra, Sutlej, Ghagra, Gandak, and Kosi. Ganga and Yamuna are the consequent rivers.

DRAINAGE PATTERNS (7:06 PM)

- The flow characteristic of a river is called a drainage pattern.
- (a) Dendritic Pattern
- It appears like the branches of a tree or a leaf.
- This is characterized by the lack of structural control.
- This is found in the Himalayan rivers. For example, the Ganga drainage pattern.
- (b) Rectangular Pattern
- The river takes sharp turns.
- It is found in the plateau regions with high structural control.
- This is found in the case of the peninsular rivers. For example, the River Mahanadi, Godavari, Cauvery, etc. follows this drainage pattern.
- (c) Trellis pattern
- The main river flows in the rift valley and tributaries join at the right angle.
- This is possible when the river flows in the rift valley. For example, the Narmada and Tapi Rivers follow this drainage pattern.
- (d) Radial Pattern
- In this, the different rivers originate from the raised platform or mountain and flow in various directions.
- For example, from Amarkantak we have the Narmada, Son, and Hasdeo rivers originate.
- (e) Parallel Drainage
- The rivers flow parallel to each other and drain directly into the sea.
- For example, West flowing rivers of the western ghats.

MAPPING OF THE RIVERS (7:28 PM)

- (a) Indus River System
- Indus flows through three countries namely China, Pakistan, and India.
- It originates near Mansarovar Lake.
- The right bank tributaries are Shyok, Gilgit, and Kabul.

- The left bank tributaries are Jhelum, Chenab, Sutlej, Ravi, and Beas.
- Nubra is the tributary of River Shyok. It comes from the Siachen glacier.
- (b) Jhelum
- Jhelum originates from Verinag.
- It flows through the Jhelum River valley called Kashmir Valley.
- It passes through Wular Lake. It is the largest freshwater lake in India.
- Jhelum River meanders while it is flowing through Kashmir. It is the only river that meanders in the youth stage.
- In Pakistan, Jhelum merges with Chenab.
- (c) Chenab
- It originates near Bara Lacha La.
- Chenab formed after the confluence of the Chandra and Bhaga rivers.
- Chandra originates from the Chandra tal.
- Bhaga originates from the Suraj tal.
- Barashigri glacier feeds the Chander tal.
- (d) Ravi
- It originates from the Kullu Hills.
- In ancient times it was called the Parushni River.
- It forms the natural boundary between India and Pakistan along the northern part of Punjab.
- (e) Beas
- It originates from the Rohtang.
- It lies completely inside India.
- It merges with Sutlej in Punjab.
- (f) Sutlej
- It originates from Lake Rakas. Its water is salty.
- Sutlej and Beas merge in Harikke.

THE TOPIC FOR THE NEXT CLASS: DRAINAGE SYSTEM OF INDIA (CONTINUED)
Geography Class 39

A BRIEF OVERVIEW OF THE CLASS (05:08 PM)
DRAINAGE SYSTEM (05:15 PM)

- **River Basins** - River channels and tributaries. How much water is available
- **Watershed** is the land part of the basin. Here quality of the land is checked. Land degradation is happening or not?. The watershed is the unit in which we try to conserve water. Watershed is referring to smaller parts.
- **Water divide**- The higher elevated part where river basins or watersheds of two rivers are differentiated

GANGA RIVER SYSTEM (05:26 PM)

- Bhagirathee and Alakananda merge together at Devprayag and the name given is Ganga
- It flows through Uttarakhand, Uttar Pradesh, Bihar, Jharkhand, and West Bengal.
- **Left bank tributary of Ganga**
- a) Ramganga- It comes from Uttarakhand. In UP it merges with Ganga. It flows through Jim Corbett national park.
- b) Gomti- Lucknow is situated near the Gomti River. It originates in UP and ends in UP. It joins Ganga in UP.

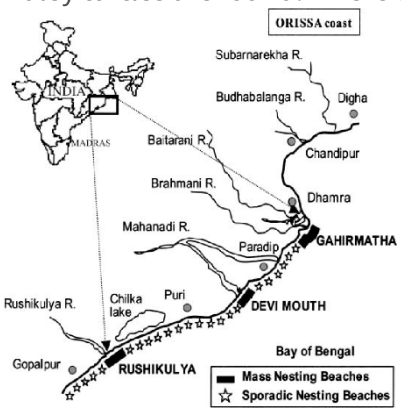
- c) Ghagra- It is an Antecedent River. It originates from the same place where the Indu, Sutlej originates i.e. Near to Mansarovar. Ghagra Merges with Ganga in Bihar. Ghagra has two tributaries i.e. Rapti and Sarda (Also known as Kali). Sarda flows between the border of India and Nepal. Ghagra is also known as the **Saryu River** in Ayodhya
- d) Gandak- It is also an Antecedent river. Gandak is also called as Narayani River.
- e) Burhi Gandak-
- f) Kosi or Sapta Kosi- It is also known as Sorrow of Bihar
- g) Mahananda- It comes from Bengal and joins Ganga.
- **Mnemonic**- Ram jane Gomti ka Ghagra Ganda Kaise
- **Right bank tributary**
- 1) Yamuna- It originates from Yamunotri. The glacier is Banderpunch. Yamuna flows through Uttarakhand, Himachal, Haryana, UP, and Delhi NCR. Yamuna merges with Ganga at Prayagraj. Yamuna only has the Right bank tributaries such as Chambal, Sindh, Betwa, and Ken
- Chambal has a tributary called Banas. Banas comes from Aravali, Rajasthan.
- Betwa and Ken are going to be interlinked under the Ken-Betwa River Interlinkage program. Surplus water from Ken River will be diverted to the Deficit Betwa River. It will submerge areas of the Panna Biosphere Reserve.
- 2) From Amarkantak, another river originates named Son River which flows parallel to Vindhya and Kaimur hills. Ganga and Son merges at Patliputra or Patna.
- Farraka Barrage is constructed over the Ganga before it enters Bangladesh. From this Barrage, one branch gets separated and flows to West Bengal. This is called the Distributary of Ganga. The name of the river is Hooghly.
- Hooghly flows through Kolkata and it has two tributaries named Ajay and Damodar Rivers. Damodar was earlier called the sorrow of Bengal. Through Damodar Valley Project (DVC) the water was controlled

BRAHMAPUTRA RIVER SYSTEM (06:00 PM)

- It originates from Chemyangdung and Angsi Glacier.
- **It has multiple names-**
- In Tibet it is calle Yarlung Tsangpo,
- In India it has three names- When it is flowing through the Himalayas of Arunachal it is called Siang, In lower parts it is called Dihang, In Assam it is called Brahmaputra.
- Once it enters Bangladesh it becomes Jamuna
- After merging with Ganga its name changes to Padma.
- Padma combines with a few other rivers to become Meghna.
- **Rightbank Tributary**
- a) Subansiri- From Arunachal, It has one smaller tributary called as Ranganadi. It is famous because it has India's largest riverine island called Majuli Island.
- b) Kameng or Jia Bhoreli- In Arunachal, it is called as Kameng and once it enters Assam it becomes Jia Bhoreli. Pakhui and Orang Tiger reserves are on either side of the Kameng river
- c) Manas- It surrounds Manas Tiger Reserve. It is also Biosphere Reserve and UNESCO heritage site. It comes from Tibet and enters Bhutan and passes Assam.
- d) Sankosh- Sankosh merges with Brahmaputra in Bangladesh
- e) Teesta- It comes from Sikkim. From Sikkim, it enters West Bengal and then goes to Bangladesh and merges with Jamuna (Brahmaputra)

- **Left Bank tributaries**
- 1) Dibang-
- 2) Lohit- At Sadiya Lohit merges with Brahamaputra. At this point, India's longest Bridge is constructed which is named Bhupen-Hazarika Bridge.
- 3) Dhansiri-
- 4) Koppili-
- 5) Barak- It originates from Manipur. Barak River is called the Surma River in Bangladesh.

PENINSULAR RIVER SYSTEM (06:21 PM)

- **Subarnrekha River**- here gold placer deposits are found
- **Baitarni and Brahmani**- There are huge mangrove areas located at Baitarni and Brahmani river delta called Bitharkanika Mangrove. It has a national park and it has a rich biodiversity. Near this, there is a beach called Gahirmatha Beach where olive ridley turtles are found. This is the largest turtle nesting site.
- 

- **Mahanadi river system**
- It was earlier called as Sorrow of Orissa. Hirakud is the largest earthen dam constructed on the Mahanadi river
- It originates from Dandkaranya.
- Tributaries are- Ib, Mand, Hasdeo, Ong, Jonk, Telen.
- **Vamsdhara river**
- It is a disputed river between Orissa and Andhra Pradesh
- **KRISHNA AND GODAVARI RIVER SYSTEM (06:53 PM)**
- It originates from Triembakeswar and Mahabaleshwar is the origin of the Krishna river
- Godavari flows from Maharashtra, Andhra Pradesh, and Telangana
- Krishna River flows from Maharashtra, Karnataka, Andhra Pradesh and Telangana
- **Tributaries of Godavari**
- **Left bank tributaries**
- Three River merges together i.e. Penganga, Wenganga, and Wardha and become Pranhita, and then it merges into Godavari.
- One from Orissa i.e. Indrawati River joins the Godavari. It has a waterfall named Chitrakoot waterfall. It is also called the "Niagra of India"
- One other river coming from Orissa i.e. Sabri and joins the Godavari in Andhra Pradesh
- **Right bank tributary**
- Manjira flows through the northernmost portion of Karnataka
- **Tributaries of Krishna**

- **Left Bank tributary**
- Bhima- Bhima basin is known for Uranium deposits
- Musi- Hyderabad is situated near this river
- **Right bank tributaries**
- Koyna- It is known for river-induced seismicity
- Ghataprabha
- Malaprabha
- Tungabhadra- It is made up of two rivers i.e Tunga and Bhadra. Hampi is located near this river.
- Note- Between Krishna and Tungabhadra there is a doab named Raichur doab.
- Note- Kolleru Lake is situated between Krishna and Godavari Lakes.
- **RIVERS BETWEEN KRISHNA AND KAVERI (07:12 PM)**
- Pennuru and Palar- Both originates from Nandi Hills
- Gandikota canyon is on Pennuru river
- **KAVERI RIVER SYSTEM**
- The river flows through Karnataka and Tamilnadu and one UT i.e. Puducherry. However, Kerala also has some portions of tributaries
- Origin is **Talkaveri** which is part of **Brahamagiri Hills** (Coorg).
- In Karnataka it flows through the plateau region. A dam was constructed i.e. Krishnaraj Sagar dam on the Kaveri river which was planned by M Visevesawaraya. This dam is the heart of the Kaveri dispute.
- Its water is completely utilized.
- Kaveri makes the most fertile region in Tamilnadu, here **Chola Empire** flourished.
- **Tributaries of kaveri**
- Left Bank tributary- Harangi, Hemawati, Shimsa, Arkawati
- Right Bank tributaries- Kabini, Bhawani, Amrawati
- Kabini- It comes from Kerala and joins Kaveri in Karnataka. There is a rich thick forest. Black leopards are found in these areas.
- **VAIGAI RIVER**
- Madurai is situated near the Vaigai River.
- Vaigai drains into Palk strait.

WEST FLOWING RIVERS (07:27 PM)

- **LUNI RIVER**
- It is called the inland- drainage river. It
- It crosses the Tropic of Cancer twice.
- It drains into the Gulf of Khambat
- **SABARMATI**
- It comes from Aravali. It drains into the Gulf of Khambat
- **NARMADA**
- It is an example of a trellis drainage pattern
- It drains into the Gulf of Khambat
- **TAPI**
- It originates from the Betul region of Madhya Pradesh.
- It drains into the Gulf of Khambat
- **MANDOVI/ MAHADEYI RIVER**
- In Goa it is called Mandovi River and in Karnataka, it is called Mahadeyi River.
- **SHARAVATI RIVER**
- Here highest waterfall is located i.e. Jog waterfall.

- **PERIYAR RIVER**
- It is famous for the Mulla-Periyar dam. It is a dam disputed between Kerala and Tamilnadu
- **PAMBA RIVER**
- It is associated with the Sabarimala temple. It flows nearby Sabarimala temple

CLIMATE OF INDIA (07:39 PM)

- **Factors affecting the Climate of India**
- a) Latitudinal Extent- The difference between the Southernmost and Northernmost latitudes which is 30 degrees. The overall climatic condition of tropical monsoons varies at the local levels due to the large latitudinal extent.
- b) Location- At the north of the Indian Ocean with a major part of the peninsular region surrounded by ocean resulting moderating effect.
- c) Presence of the Himalayas- It protects India from the cold and dry winds of Central Asia during winters. It also influences monsoons in India.
- d) Other topographical features- Such as the Western Ghats being perpendicular to Monsoon winds receiving high precipitation on the western side and rainshadow along the eastern side. The orientation of Aravali parallel to Monsoon winds resulting very low precipitation in Rajasthan
- e) Monsoon winds- The most dominant factor of Indian climate due to which it is called as Monsoon climate.
- f) Jet stream- Subtropical westerly jetstreams, Tropical easterly jetstreams, and Somali Jetstreams influence winter and summer conditions over India.
- g) Cyclones- The coastal regions of Tamilnadu, Andhra Pradesh, Orissa, and West Bengal are largely affected by Tropical cyclones.
- h) El-Nino, La-Nina, Maden-Julian oscillation, Indian Ocean Dipole, etc influence rainfall over India.

MONSOON (07:53 PM)

- The word Monsoon is derived from the Arabic word "Mausim" which means a seasonal reversal of wind.
- **Classical theory of the origin**
- This explains the phenomenon of Monsoons as large-scale Sea breezes and Land breezes due to the reversal of temperature and pressure conditions from summer to winter in the Northern Plains.
- During summers, high temperatures and low pressure attract winds from the Arabian Sea.
- During winters, Low temperatures and high pressures cause winds to blow away from the plains towards the sea.

The topic for the next class:- The Modern Theory of Monsoon

Geography Class 40

A BRIEF OVERVIEW OF THE CLASS (05:12 PM)

MODERN THEORY OF MONSOON (05:24 PM)

- **Winter season (November- February)**
- Low temperature and high-pressure conditions are prevailing over the northern plains.
- It causes Anti-cyclonic circulations causing winds to blow outward from the land over which air is subsiding.
- The sky is clear and dry.